Chapter Five: Decentralized Stablecoins

The prices of cryptocurrencies are extremely volatile. To mitigate this volatility, stablecoins that are pegged to other stable assets such as the USD were created. Stablecoins help users to hedge against this price volatility and was created to be a reliable medium of exchange. Stablecoins have since quickly evolved to be a strong component of DeFi that is pivotal to this modular ecosystem.

There are 19 stablecoins currently listed on CoinGecko. The top 5 stablecoins has a market capitalization totaling over $5 billion.

Top 5 Cryptocurrency Stablecoins (Feb 2020)		
Rank	**Bank**	**Market Cap. ($ million)**
1	Tether (USDT)	4,284
2	USD Coin (USDC)	443
3	Paxos Standard (PAX)	202
4	True USD (TUSD)	142
10	Dai (DAI)	123

Source: CoinGecko.com

We will be looking into USD-pegged stablecoins in this chapter. Not all stablecoins are the same as they employ different mechanisms to keep their peg against USD. There are two types of pegs, namely fiat-collateralized and crypto-collateralized. Most stablecoins employ the fiat-collateralized system to maintain their USD peg.

For simplicity, we will look at two USD stablecoins, Tether (USDT) and Dai (DAI) to showcase the differences in their pegging management.

Tether (USDT) pegs itself to $1 by maintaining reserves of $1 per Tether token minted. While Tether is the largest and most widely used USD stablecoin with daily trading volumes averaging approximately $30 billion in the month of January 2020, Tether reserves are kept in financial institutions and users will have to trust Tether as an entity to actually have the reserve amounts that they claim. Tether is therefore a **centralized, fiat-collateralized stablecoin**.

Dai (DAI) on the other hand, is collateralized using cryptocurrencies such as Ethereum (ETH). Its value is pegged to $1 through protocols voted on by a decentralized autonomous organization and smart contracts. At any given time, the collateral to generate DAI can be easily validated by users. DAI is a **decentralized, crypto-collateralized stablecoin**.

Based on the top 5 stablecoins' market capitalization, Tether dominates the stablecoin market with approximately 80% of market share. Although DAI's market share only stands at about 3%, its trading volume has been increasing at a much faster rate. DAI's trading volume increased by over 4,000% relative to Tether's growth of 126% since the start of January 2020.

DAI is the native stablecoin used most widely in the DeFi ecosystem. It is the preferred USD stablecoin used in DeFi trading, lending and more. To understand DAI further, we will introduce you to its platform, Maker.

Maker

What is Maker?

MAKER

Maker is a smart-contract platform that runs on the Ethereum blockchain and **has three tokens**: stablecoins, Sai and Dai (both algorithmically pegged to $1), as well as its governance token, Maker (MKR).

Sai (SAI) is also known as Single Collateral Dai and is backed only by Ether (ETH) as collateral.

Dai (DAI) was launched in November 2019 and is also known as Multi-Collateral Dai. It is currently backed by Ether (ETH) and Basic Attention Token (BAT) as collaterals with plans to add other assets as collaterals in the future.

Maker (MKR) is Maker's governance token and users can use it to vote for improvements on the Maker platform via the Maker Improvement Proposals. Maker is a type of organization known as a Decentralized Autonomous Organization (DAO). We will look further into this under the governance subsection.

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What are the Differences between Sai and Dai?

Maker initially started out on 19 December 2017 with the Single Collateral Dai. It was minted using Ether (ETH) as the sole collateral. On 18 November 2019, Maker announced the launch of the new Multi-Collateral Dai, which can be minted using either Ether (ETH) and/or Basic Attention Token (BAT) as collateral, with plans to allow other cryptocurrencies to back it in the future. To reiterate,

Single-Collateral Dai = Legacy Dai = Sai

Multi-Collateral Dai = New Dai = Dai

Moving forward, Multi-Collateral Dai will be the de-facto stablecoin standard maintained by Maker and eventually, SAI will be phased out and no longer supported by Maker. For brevity, we will only be using Multi-Collateral Dai (DAI) to walk through examples in the following sections.

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How does Maker Govern the System?

Recall our brief mention on Decentralized Autonomous Organization (DAO)? That's where the Maker (MKR) token comes in. MKR holders have voting rights proportional to the amount of MKR tokens they own in the DAO and can vote on parameters governing the Maker Protocol.

The parameters that the MKR holders vote on are vital in keeping the ecosystem healthy, which in turn helps ensure that Dai remains pegged to $1. We will briefly go through three key parameters which you will need to know in the Dai stablecoin ecosystem:

I. Collateral Ratio

The amount of Dai that can be minted is dependent on the collateral ratio.

Ether (ETH) collateral ratio	=	150%
Basic Attention Token (BAT) collateral ratio	=	150%

Essentially, what it means with a collateral ratio of 150% is that to mint $100, you need to deposit a minimum of $150 worth of ETH or BAT.

II. Stability Fee

It is equivalent to the 'interest rate' which you are required to pay along with the principal debt of the vault. The stability fee is at 8% as of February 2020.

III. Dai Savings Rate (DSR)

The Dai Savings Rate (DSR) is the interest earned by holding Dai over time. It also acts as a monetary tool to influence the demand of Dai. The DSR rate is set at 7.50% as of February 2020.

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Motivations to Issue Dai

Why would you *want* to lock up a higher value of ETH or BAT only to issue Dai with a lower value? You could have sold your assets directly to USD instead.

There are three possible cases:

I. You need cash now and you have an asset that you believe will be worth more in the future.

- In this case, you could hold your asset in the Maker vault and get the money now by issuing Dai.

II. You need cash now but do not want to risk triggering a taxable event when selling your asset.

- Instead, you will draw the loans by issuing Dai.

III. Investment Leverage

- You are able to conduct investment leverage on your assets given that you believe the value of your assets would go up.

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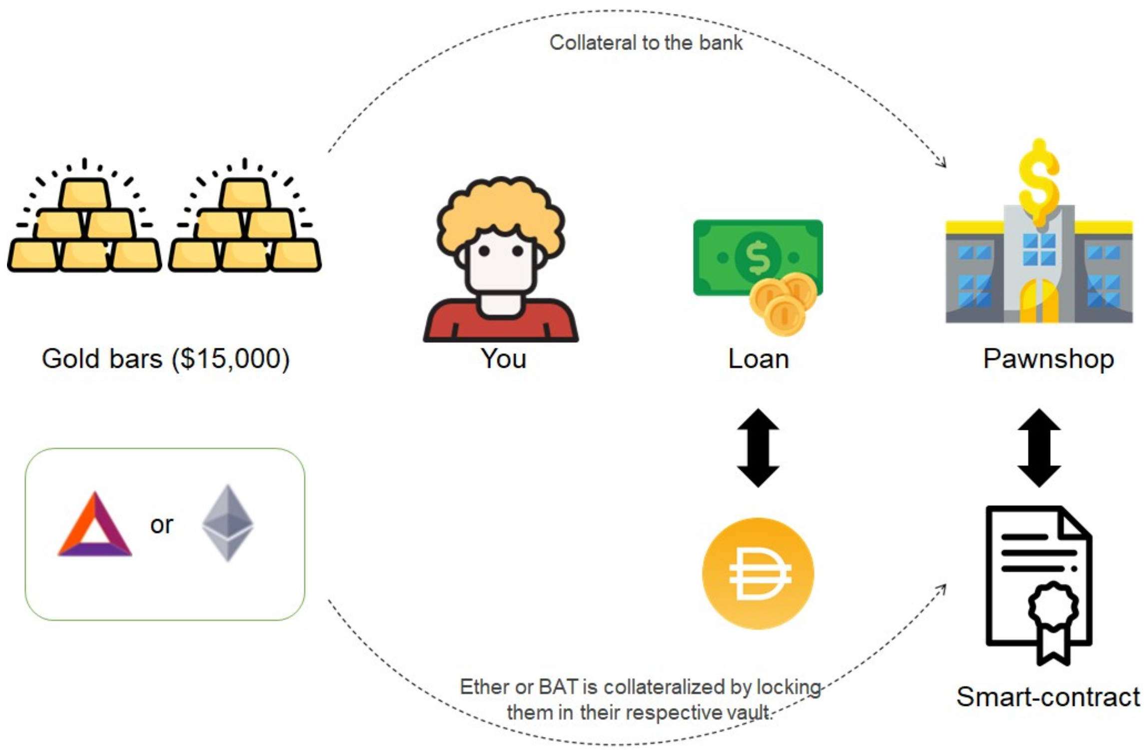
How do I get my hands on some Dai (DAI)?

There are two ways you can get your hands on some Dai (DAI):

Minting Dai

We will walk through how DAI can be minted using a pawnshop analogy.



Let’s assume that one day you are in need of $10,000 cash, but all you have are gold bars worth $15,000 at home. Believing that the price of gold will increase in the future, instead of selling the gold bars for cash, you decide to go to a pawnshop to borrow $10,000 cash by putting your gold bars as collateral for it. The pawnshop agrees to lend you $10,000 with an interest of 8% for the cash loan. Both of you sign a contract agreement to finalize the transaction.

Now let’s change the terminology to get the narrative of DAI:

Gold bars (Collateral)		Ether or Basic Attention Token (Collateral – Multi-Collateral Dai)
Cash loan		Dai (DAI)
The pawnshop		Maker

Contract agreement	➡	Smart Contract (Vault)
Loan interest	➡	Stability Fee

What happens is that you will mint or 'borrow' Dai via the Maker platform by putting your Ether (ETH) or Basic Attention Token (BAT) as collateral. You will have to repay your 'loan' along with the 'loan interest' which is the stability fee when you want to redeem your ETH or BAT at the end of your loan.

To provide an overview, let's walk through how you can mint your own Dai.

On the Maker platform (www.oasis.app), you can borrow Dai by putting your ETH or BAT into the vault. Assuming ETH is currently worth $150, you can thus lock 1 ETH into the vault and receive a maximum of 100 DAI ($100) with a 150% collateral ratio.

You should not draw out the maximum of 100 DAI that you are allowed to but leave some buffer in the event that ETH price decreases. It is advisable to give a wider gap to ensure your collateral ratio always remains above 150%. This ensures that your vault will not be liquidated and charged the 13% liquidation penalty in the event that ETH falls in price and your collateral ratio falls below 150%.

Trading Dai

The above methods are all the ways DAI are created. Once DAI is created, you can send it anywhere you want. Some users may send their DAI to cryptocurrency exchanges and you may also buy DAI from these secondary markets without the need to mint them.

Buying DAI this way is easier as you don't have to lock up collateral and do not have to worry about the collateral ratio and stability fee.

We will keep this section brief - you can check out CoinGecko for the list of exchanges that trades DAI.

~

Black Swan Event

A black swan event is an unpredictable and extreme event that may cause severe consequences. In the case where both ETH and BAT has a significant drop in price, Emergency Shutdown is triggered. It is a process used as a last resort to settle the Maker Platform by shutting the system down. The process is to ensure the holders of Dai holders and Vault users receive the net value of assets they are entitled to.

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Why Use Maker?

As previously mentioned in Section 2: Stablecoins, there are many stablecoins out there and the core distinctions of these coins lie in their protocol. Unlike most stablecoin platforms, Maker is **fully** operating on the distributed ledger. Thus, Maker inherently possesses the characteristics of the blockchain: secured, immutable and most importantly, transparent. Additionally, Maker's infrastructures have strengthened the security of the system with comprehensive risk protocols and mechanisms via real-time information.

And that's it for Makers' Stablecoin, Dai—if you're keen to get started or test it out, we've included step-by-step guides on how to (i) mint some DAI for yourself and (ii) save DAI to earn interest. Otherwise, head on to the next section to read more on the next DeFi app!

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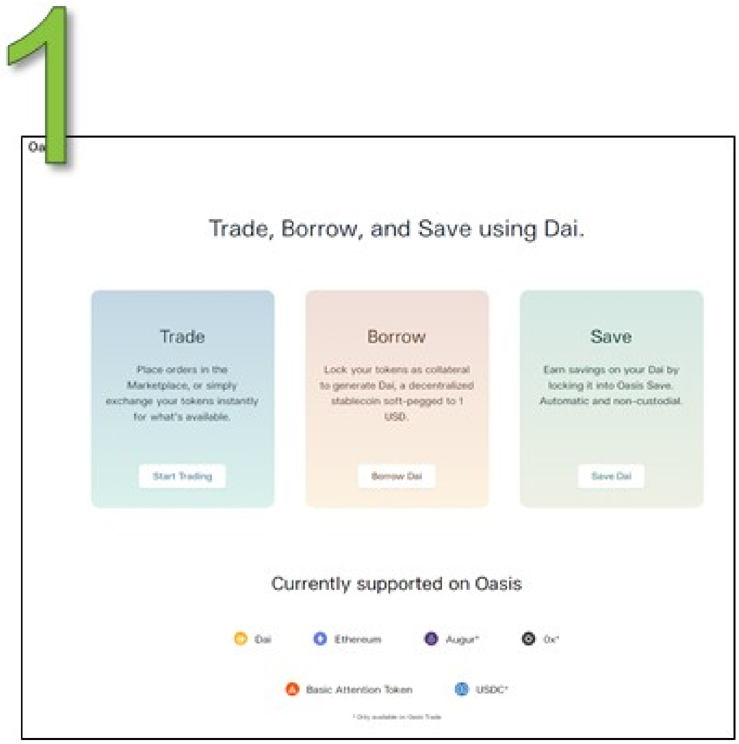
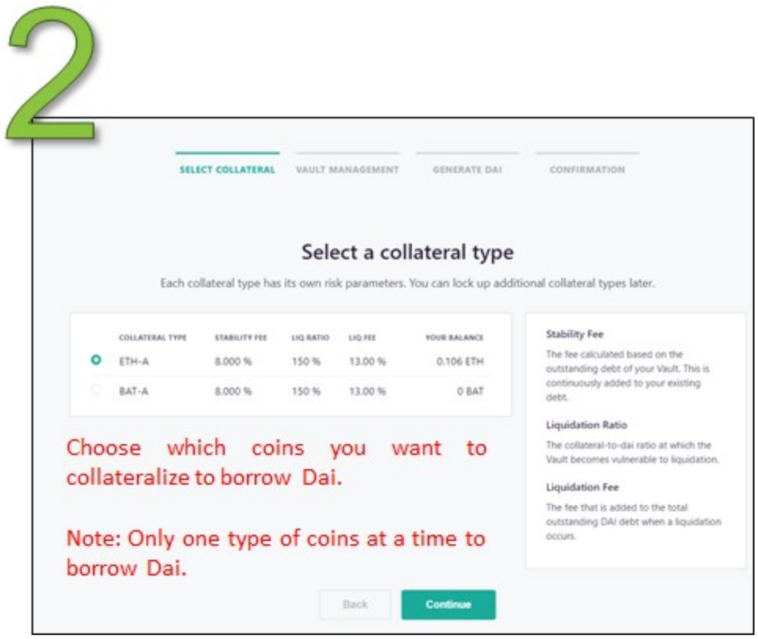
Maker: Step-by-Step Guide
Minting your own DAI



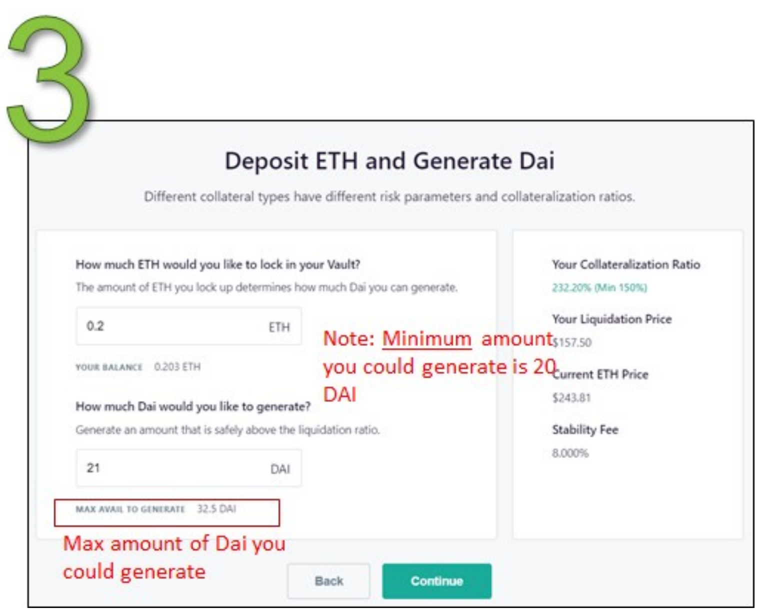
Step 1

- Go to https://oasis.app/
- Click Borrow
- You will be asked to connect your wallet. Connecting your wallet is free, all you need to do is sign a transaction



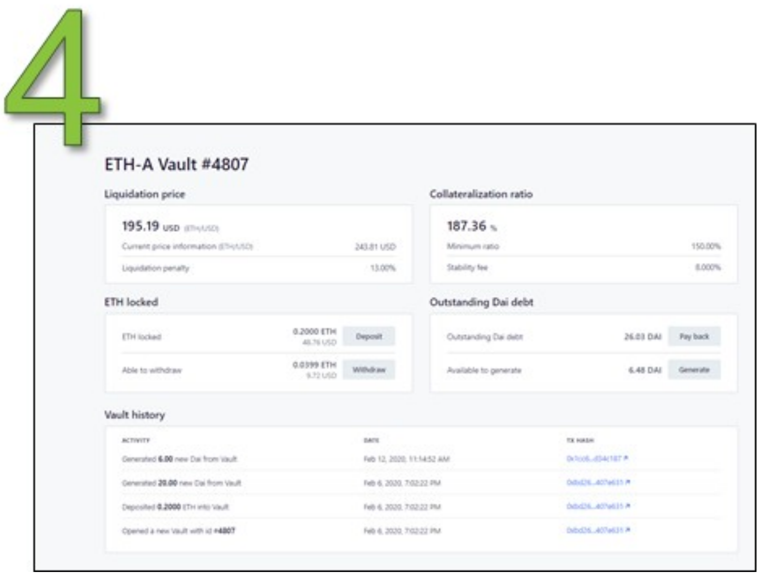
Step 2

- Click "Start Borrow" once you are on the borrow page (https://oasis.app/borrow/)
- Choose which cryptoasset you want to lock-in (collateralize)



Step 3

- In this example, I chose to lock-in ETH
- Insert any amount you wish to lock. Note: You have to lock an equal amount of 20 DAI and above. (ie: You can't borrow DAI less than 20 DAI)
- Click Continue and follow the instructions afterward



Step 4

- Congratulations! Your ETH vault is now created!

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In addition to minting DAI, you can also save on Maker's platform to earn interest on your assets. We've prepared a step-by-step guide on how to save your DAI below:

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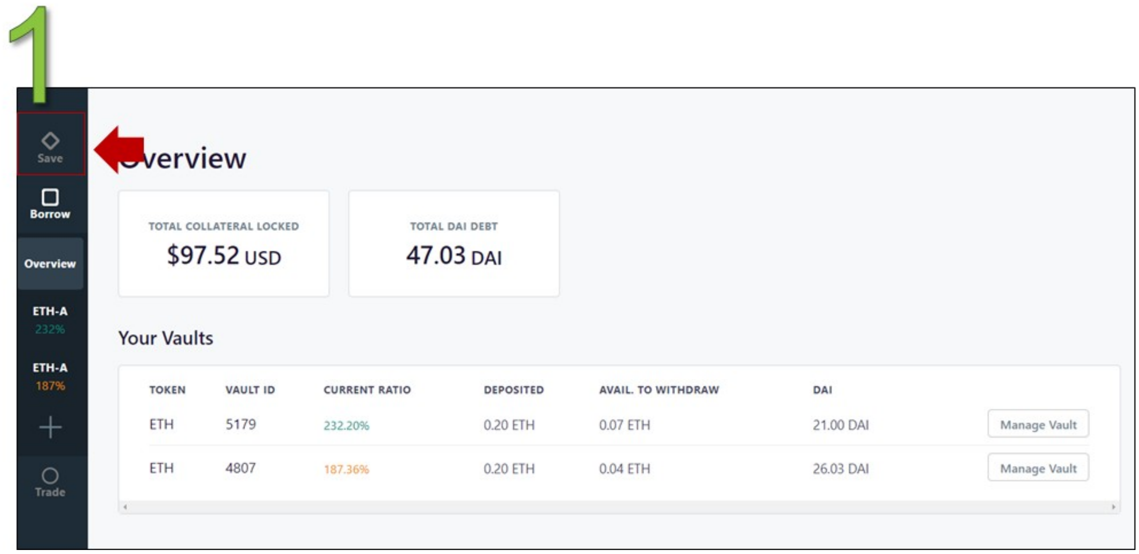
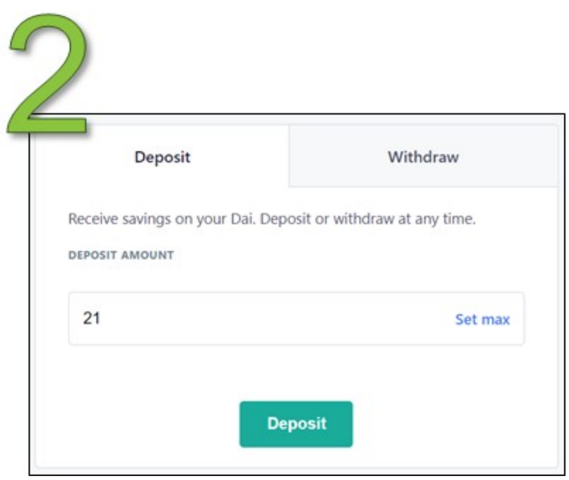
Saving your DAI



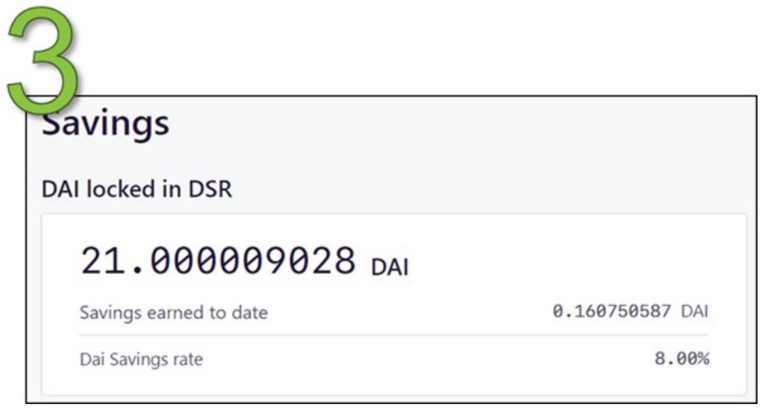
Step 1

- Navigate to the Save page on the left sidebar (https://oasis.app/save)



Step 2

- Click Start Save
- Enter the amount of DAI you wish to save
- Click deposit
- Confirm in wallet



Step 3

- And it's done!
- Note: You only have one DSR account. If you were to deposit more DAI after your first deposit, it will be added to it.

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Recommended Readings

1. Maker Protocol 101 (Maker) https://docs.makerdao.com/maker-protocol-101
2. Maker for Dummies: A Plain English Explanation of the Dai Stablecoin (Gregory DiPrisco) https://medium.com/cryptolinks/maker-for-dummies-a-plain-english-explanation-of-the-dai-stablecoin-e4481d79b90
3. What's MakerDAO and what's going on with it? Explained with pictures. (Kerman Kohli) https://hackernoon.com/whats-makerdao-and-what-s-going-on-with-it-explained-with-pictures-f7ebf774e9c2
4. How to get a DAI saving account (Ryan Sean Adams) https://bankless.substack.com/p/how-to-get-a-dai-saving-account

Chapter Six: Decentralized Lending and Borrowing

One of the most common services offered by the financial industry is the lending and borrowing of funds, which was made possible by the concept of credit and collateralization.

It can be argued that the invention of commercial-scale lending and borrowing was what brought about the Renaissance age as the possibility for the less wealthy to acquire startup funds led to a flurry of economic activity. Thus, the economy began to grow at an unprecedented pace.[9]

Entrepreneurs can borrow the upfront capital needed to establish a business by collateralizing the business while families can get a mortgage for a house that would otherwise be too costly to buy in cash, whilst using the house as collateral. On the other hand, the wealth accumulated can be lent off as capital to lenders. This system reduces the risk of borrowers absconding with the borrowed funds.

However, this system requires some form of trust and an intermediary. The role of an intermediary is taken up by banks and trust is maintained via a convoluted system of credit, whereby the borrower must exhibit the ability to repay the loan in order to be qualified to borrow, among a laundry list of other qualifications and requirements by the banks.

This has led to various challenges and shortfalls of the current lending and borrowing system, such as restrictive funding criteria, geographical or legal restriction to access banks, high barriers to loan acceptance, and the exclusivity of only the wealthy to enjoy the benefits of low-risk high-returns lending.

In the DeFi landscape, such barriers do not exist as banks are no longer necessary. With enough collateral, anyone can have access to capital to do whatever they want. Capital lending is also something that is no longer enjoyed only by the wealthy, everyone can contribute to a decentralized liquidity pool of which borrowers can take from and pay back at an algorithmically-determined interest rate. In contrast to applying for a loan from the bank where there are stringent Know-your-customer (KYC) and Anti-money laundering (AML) policies, one only needs to provide collateral to take a loan in DeFi.

We will explore just how such bankless lending and borrowing mechanism is possible with Compound Finance, a Defi lending and borrowing protocol.

Compound

Compound Finance is an Ethereum-based open-source money market protocol where anyone can **supply or borrow cryptocurrencies** frictionlessly. As of Feb 2020, 7 different tokens—Basic Attention Token (BAT), Dai (DAI), Ether (ETH), Augur (REP), USD Coin (USDC), Wrapped Bitcoin (WBTC) and 0x (ZRX)—can be supplied or used as collateral on the Compound Platform.

Compound operates as a liquidity pool that is built on the Ethereum blockchain. Suppliers supply assets to the pool and earn interest, while borrowers take a loan from the pool and pay interest on their debt. In essence, Compound bridges the gaps between lenders who wish to accrue interest from idle funds and borrowers who wish to borrow funds for productive or investment use.

In Compound, interest rates are denoted in Annual Percentage Yield (APY) and differ between assets. Compound derives the interest rates for different assets through algorithms which account for supply and demand of the asset.

Essentially, Compound lowers the friction for lending/borrowing by allowing suppliers/borrowers to interact directly with the protocol for interest rates without needing to negotiate loan terms (eg. maturity, interest rate, counterparty, collaterals), thereby creating a more efficient money market.

~

How much interest will you receive, or pay?

The Annual Percentage Yield (APY) differs between assets as it is algorithmically set based on the supply and demand of the asset. Generally, the higher the borrowing demand, the higher the interest rate (APY) and vice versa.

All Markets

Market	Gross Supply	Supply APY	Gross Borrow	Borrow APY
Ether ETH	$98.42M +3.02%	0.01% –	$770k +1.17%	2.10% –
Dai DAI	$32.57M -0.95%	7.58% –	$17.50M +1.39%	8.00% –
USD Coin USDC	$26.86M -0.80%	5.48% +0.36	$17.65M +3.00%	8.92% +0.26

Using the DAI stablecoin as an example, a lender would earn 7.58% (as of Feb 2020) in a year while a borrower would be paying 8.00% interest after a year.

~

Do I need to register for an account to start using Compound?

No, you do not need to register for an account and that's the beauty of Decentralized Finance applications! Unlike traditional financial applications where users are required to go through lengthy processes to get started, Compound users do not need to register for anything.

Anyone with a supported cryptocurrency wallet such as Argent and Metamask can start using Compound immediately.

~

Start earning interest on Compound

To earn interest, you will have to supply assets to the protocol. As of February 2020, Compound accepts 7 types of tokens.

Once you have deposited your asset into Compound, you will immediately begin to earn interest on the assets you have put in! Interest accrued on the amount that you have supplied and is calculated after each Ethereum block (average ~13 seconds).

Upon deposit, you will receive corresponding amounts of cTokens. If you supply DAI, you will receive cDAI, if you supply Ether, you will receive cETH, and so on). Interest is not immediately distributed to you, but rather accrues on the cTokens which you now hold and are redeemable for the underlying asset and interest it represents.

~

cTokens?

cTokens represent your balance in the protocol and accrue interest over time. In Compound, interest earned is not distributed immediately but is instead accrued in cTokens.

Let's go through this with an **example.** Assume that you have supplied 1,000 DAI on 1 January 2019 and APY has been constant at 10.00% throughout 2019.

On 1 January 2019, after you have deposited 1,000 DAI, you will be given 1,000 cDAI. In this case, the exchange rate between DAI and cDAI is 1:1.

On 1 January 2020, after 1 year, your 1,000 cDAI will now increase in value by 10%. The new exchange rate between DAI and cDAI is 1:1.1. Your 1,000 cDAI is now redeemable for 1,100 DAI.

1 Jan 2019: Deposit 1,000 DAI. Receive 1,000 cDAI. Exchange Rate: 1 cDAI = 1 DAI **1 Jan 2020:** Redeem 1,000 cDAI. Receive 1,100 DAI. Exchange Rate: 1 cDAI = 1.1 DAI (cDAI value increased by 10%)

To account for the interest accrued, cTokens become convertible into an increasing amount of the underlying asset it represents over time. cTokens are also ERC-20 tokens, meaning you can easily transfer the "ownership" of supplied assets if someone wants to take over your position as a supplier.

~

Start borrowing on Compound

Before borrowing, you have to supply assets into the system as collateral for your loan. Each asset has a different collateral factor. The more assets you supply, the greater is your borrowing power.

Borrowed assets are sent directly to your Ethereum wallet and from there, you can use them as you would any cryptoasset—anything you want to! Do note that that borrowing incurs a small fee of 0.025% to avoid spams and misuse of the Compound protocol.

~

Price movement of collateral asset

If you're thinking about putting in collateral to take a loan, you may be wondering - what happens if the value of the collateral changes? Let's see:

1. Collateral value moves up

    If the value of the asset you used as collateral goes up, your collateral ratio also goes up, which is fine - nothing will happen and you can draw a bigger loan if you'd like to.

2. Collateral value moves down

    On the other hand, if the collateral goes down such that your collateral ratio is now below the required collateral ratio, your collateral will be partially sold off along with a 5% liquidation fee. The process of selling off your collateral so that you achieve the minimum collateral ratio is known as liquidation.

~

Liquidation

Liquidation occurs when the value of the collateral provided is less than the borrowed funds. This is to ensure that there is always excess liquidity for withdrawal and borrowing of funds, meanwhile protect lenders against default risk. The current liquidation fee is 5%.

And that's it for Compound - if you're keen to get started or test it out, we've included step-by-step guides on how to (i) supply funds to the pool and

(ii) borrow from the pool. Otherwise, head on to the next section to read more on the next DeFi app!

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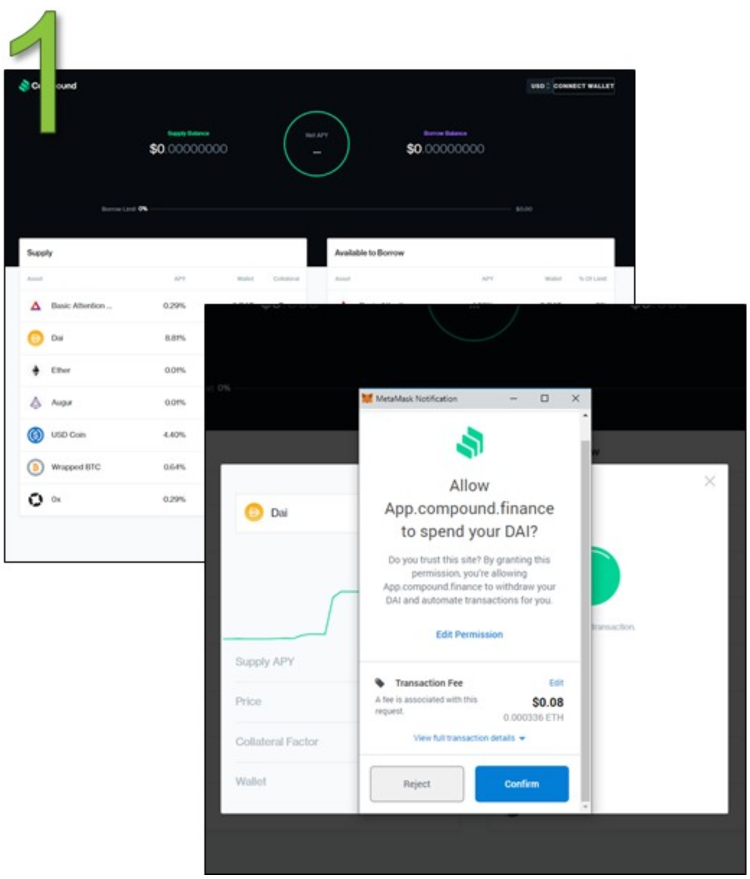
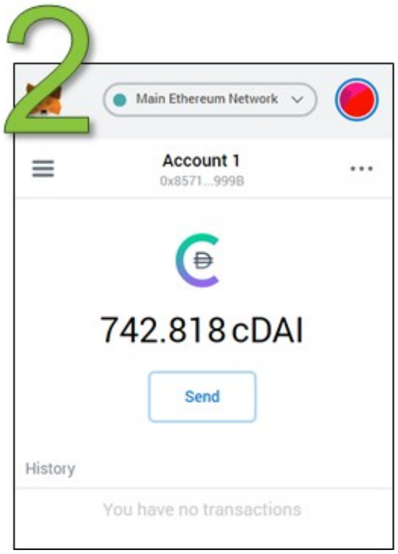
Compound Finance: Step-by-Step Guides
Supplying fund to the pool



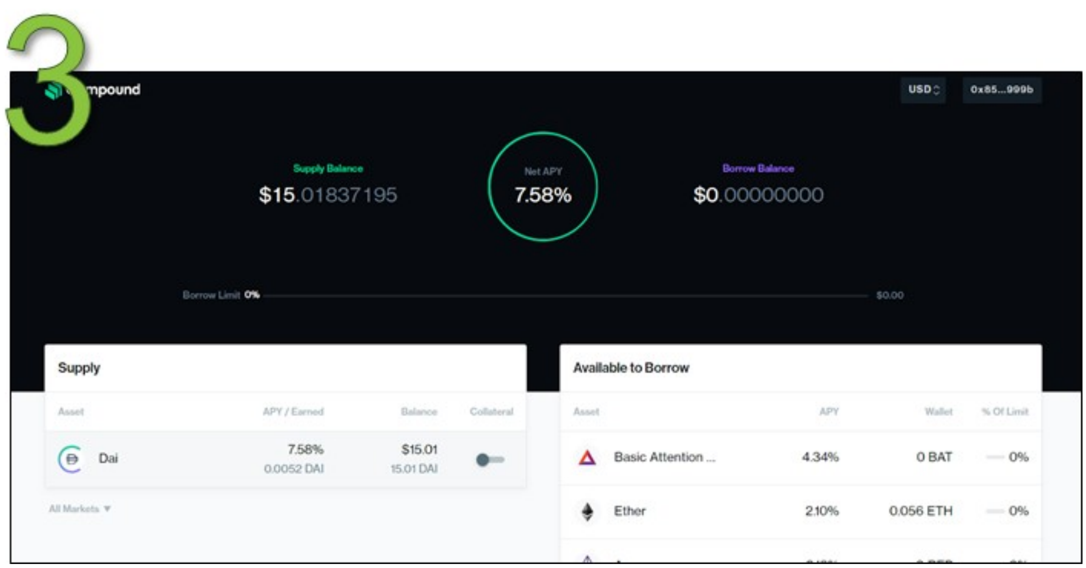
Step 1

- Head over to https://app.compound.finance
- Connect your wallet. Follow your wallet's instructions
- Deposit cryptocurrencies into the liquidity pool (any of the 7 tokens)



Step 2

- Receive cTokens
- When you sign up for a fixed deposit, the bank will issue a fixed deposit certificate as proof of placement. Similarly, after supplying assets, you will get cTokens which represent the type and amount of assets you have deposited
- The cTokens act as a claim of deposit and record the interest you earn. Likewise, you have to use it to redeem or withdraw your assets



Step 3

- Earn Interest
- The moment you deposit assets, you start to earn interest. By holding the cTokens, the interest will accrue on your account.

Step 4

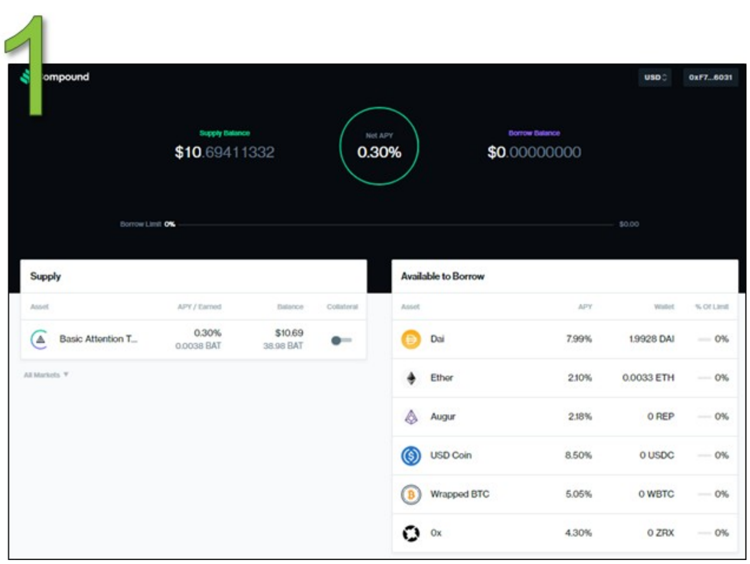
- Redeem cToken
- Over time, the interest accumulates and each cTokens is convertible into a greater value of underlying assets. You can redeem the cTokens anytime and receive the assets back to your wallet instantly.

~

Borrowing fund from the pool

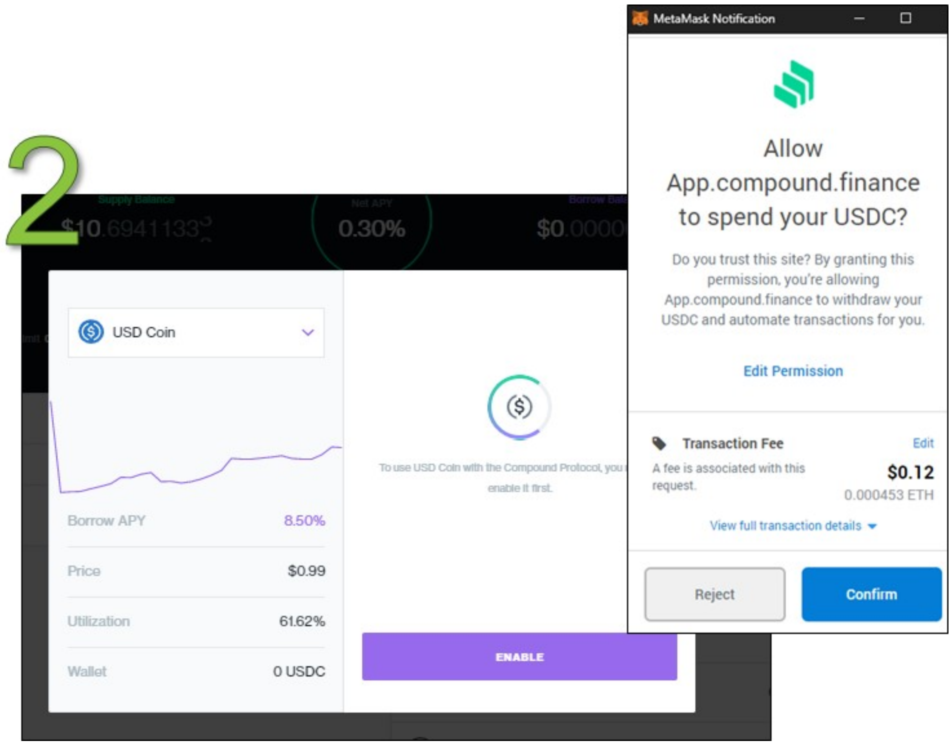
Note:

- Before you could start borrowing, you are required to supply a value worth of assets into the compound as a form of collateral.
- Each token has its own collateral factor. A collateral factor is the ratio of how much you have to supply in order to borrow.
- Another note is you cannot supply and borrow the same token at the same time.



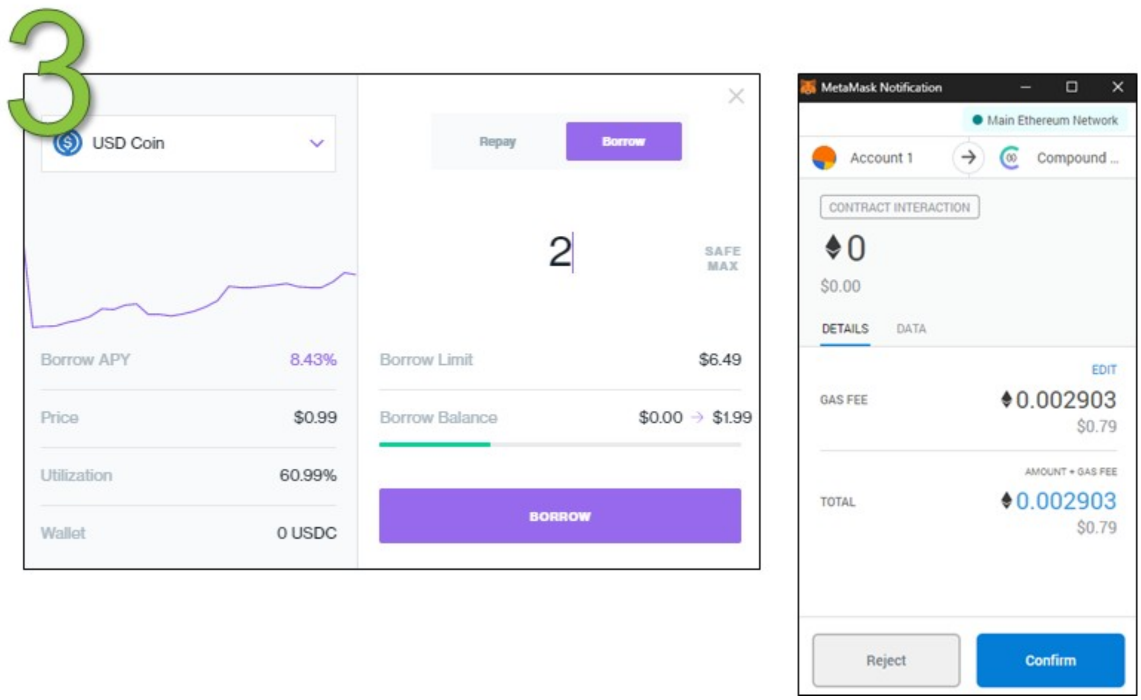
Step 1

- Go the Compound's main page https://app.compound.finance/
- Choose which tokens you wish to borrow on the right bar. I chose USDC



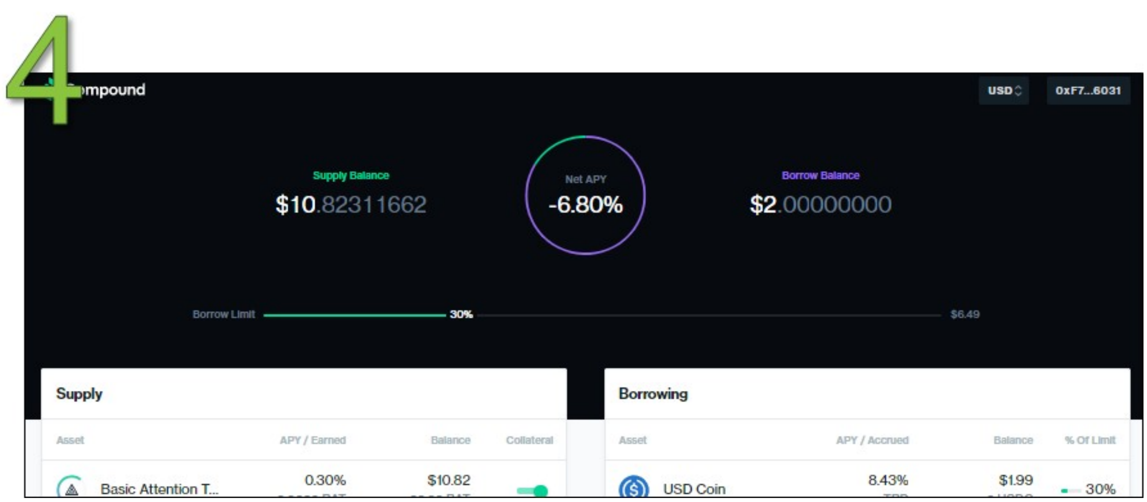
Step 2

- A pop-up on USDC will appear
- First-timers will need to enable it
- Each token has to enable it individually



Step 3

- I punched in the amount of how much I want to borrow. I wanted to borrow 2 USDC
- Confirm the transaction with your wallet



Step 4

- Finished!

- You can see how much you supply and how much you borrow on the main page.

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Recommended Readings

1. The DeFi Series – An overview of the ecosystem and major protocols (Alethio) https://medium.com/alethio/the-defi-series-an-overview-of-the-ecosystem-and-major-protocols-da27d7b11191
2. Compound FAQ (Robert Leshner) https://medium.com/compound-finance/faq-1a2636713b69
3. DeFi Series #1 - Decentralized Cryptoasset Lending & Borrowing (Binance Research) https://research.binance.com/analysis/decentralized-finance-lending-borrowing
4. Zero to DeFi – A beginner's guide to earning passive income via Compound Finance (Defi Pulse) https://defipulse.com/blog/zero-to-defi-cdai/
5. I took out a loan with cryptocurrency and didn't sign a thing (Stan Schroeder) https://mashable.com/article/defi-guide-ethereum-decentralized-finance.amp
6. Earn passive income with Compound. (DefiZap) https://defitutorials.substack.com/p/earn-passive-income-with-compound

Chapter Seven: Decentralized Exchanges (DEX)

While Centralized Exchanges (CEXs) allow for large trades to happen with plenty of liquidity, it still carries a lot of risks because users do not have ownership of their assets in exchanges. In 2019, over $290 million worth of cryptocurrencies were stolen and over 500,000 login information were leaked from exchanges.[10]

More people are realizing these risks and are turning to Decentralized Exchanges (DEXs). DEXs work by using smart contracts and on-chain transactions to reduce or eliminate the need for an intermediary. Some popular Decentralized Exchanges include projects like Kyber Network, Uniswap, Dex Blue and dYdX.

There are two kinds of DEXs - order book-based DEXs and liquidity pool-based DEXs. Order book DEXs like dYdX and dex.blue operate similarly to CEXs where users can place buy and sell orders at either their chosen limit prices or at market prices. The main difference between the two is that for CEXs, assets for the trade would be held on the exchange wallet whereas for DEXs, assets for trade can be held on users' own wallets.

However, one of the biggest problems facing order book-based DEXs is liquidity. Users may have to wait a long time for their orders to be filled in the order book. To solve this issue, liquidity pools-based DEXs were introduced. Liquidity pools are essentially reserves of tokens in smart contracts and users can buy or sell tokens instantly from the available tokens in the liquidity pool. The price of the token is determined algorithmically and increases for large trades. DEXs liquidity pools can be shared across multiple DEX platforms and this pushes up the available liquidity on any single platform. Examples of liquidity pools-based DEXs are Kyber Network, Bancor, and Uniswap. We will be going through the Uniswap example in this book.

One of the features offered by CEXs is the margin trading function. Margin trading enables an investor to trade leveraged positions, boosting one's purchasing power to gain potentially higher returns. Innovations to bring margin trading on DEXs have appeared as well. Examples of DEXs offering decentralized margin trading are dYdX, NUO Network and DDEX. In this book, we will be exploring dYdX which combines both decentralized lending and borrowing markets with margin trading on their exchange.

Uniswap

Uniswap Exchange is a decentralized token exchange protocol built on Ethereum that allows direct swapping of tokens without the need to use a centralized exchange. When using a centralized exchange, you will need to deposit tokens to an exchange, place an order on the order book, and then withdraw the swapped tokens.

On Uniswap, you can simply swap your tokens directly from your wallet without having to go through the three steps above. All you need to do is send your tokens from your wallet to Uniswap's smart contract address and you will receive your desired token in return in your wallet. There is no order book and the token exchange rate is determined algorithmically. All this is achieved via liquidity pools and the automated market maker mechanism.

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Liquidity Pools

Liquidity pools are token reserves that sit on Uniswap's smart contracts and are available for users to exchange tokens with. For example, using ETH-DAI trading pair with 100 ETH and 20,000 DAI in the liquidity reserves, a user that wants to buy ETH using DAI may send 202.02 DAI to the Uniswap smart contract to get 1 ETH in return. Once the swap has taken place, the liquidity pool is left with 99 ETH and 20,202.02 DAI.

Liquidity pools reserves are provided by liquidity providers who are incentivized to obtain a proportionate fee of Uniswap's 0.3% transaction fee. This fee is charged for every token swap on Uniswap.

There are no restrictions and anyone can be a liquidity provider - the only requirement is that one needs to provide ETH and the quoted trading token to be swapped to at the current Uniswap exchange rate. As of Feb 2020, over 125,000 ETH have been locked into Uniswap. The amount of reserves held by a pool plays a huge role in determining how prices are set by the Automated Market Maker Mechanism.

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Automated Market Maker Mechanism

Prices of assets in the pool are algorithmically determined using the Automated Market Maker (AMM) algorithm. AMM works by maintaining a Constant Product based on the amount of liquidity in both sides of the pool.

Let's continue the ETH-DAI liquidity pool example which has 100 ETH and 20,000 DAI. To calculate the Constant Product, Uniswap will multiply both these amounts together.

ETH liquidity (x)	*	DAI liquidity (y)	=	Constant Product (k)
100	*	20,000	=	2,000,000

Using AMM, at any given time, the Constant Product (k) must always remain at 2,000,000. If someone buys ETH using DAI, ETH will be removed from the liquidity pool while DAI will be added into the liquidity pool.

The price for this ETH will be determined asymptotically. The larger the order, the larger the premium that is charged. Premium refers to the additional amount of DAI required in order to purchase 1 ETH compared to the original price of 200 DAI per ETH.

The table below further elaborates the asymptotic pricing and the movement of liquidity when orders to purchase ETH are made.

ETH Purchased	*Cost per ETH in DAI*	*Total Cost in DAI*	*Premium*	*New DAI Liquidity*	*New ETH liquidity*	*Product (k)*
1	202.02	202.02	1.01%	20,202.02	99	2,000,000

5	210.52	1,052.63	5.26%	21,052.63	95	2,000,000
10	222.22	2,222.22	11.11%	22,222.22	90	2,000,000
50	400	20,000	200%	40,000	50	2,000,000
75	800	60,000	400%	80,000	25	2,000,000
99	20,000	1,980,000	10,000%	2,000,000	1	2,000,000
100	Infinity	Infinity	Infinity	Infinity	0	2,000,000

As can be seen from the table above, the larger the amount of ETH that a user wishes to buy, the larger the premium that will be charged. This ensures that the liquidity pool will never be out of liquidity.

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How to get a token added on Uniswap?

Unlike centralized exchanges, Uniswap as a decentralized exchange does not have a team or gatekeepers to evaluate and decide on which tokens to list. Instead, any ERC-20 token can be listed on Uniswap by anyone and be traded as long as liquidity exists for the given pair. All a user needs to do is to interact with the platform to register the new token and a new market will be initialized for this token.

And that's it for Uniswap - if you're keen to get started or test it out, we've included step-by-step guides on how to (i) swap tokens, (ii) provide liquidity and (iii) stop providing liquidity. Otherwise, head on to the next section to read more on the next DeFi app!

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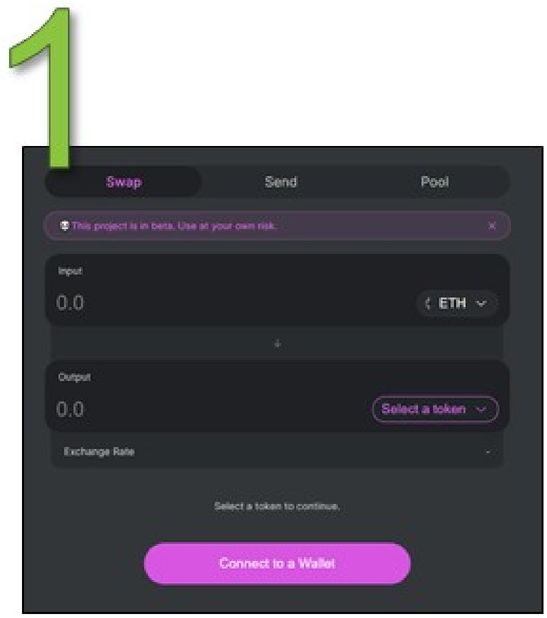
Uniswap: Step-by-Step Guide

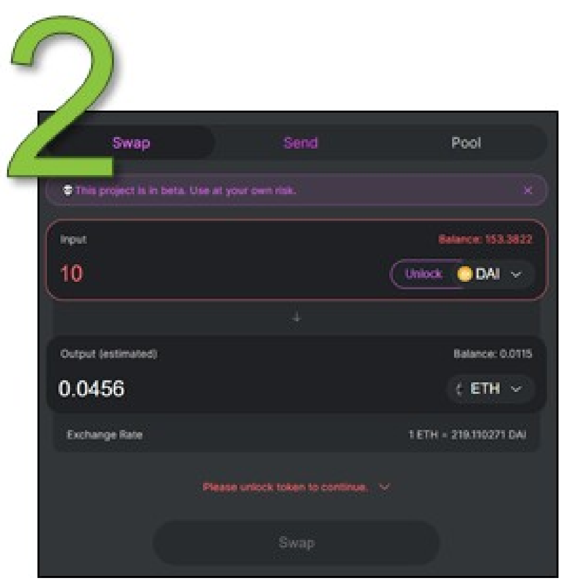
Swapping Tokens



Step 1

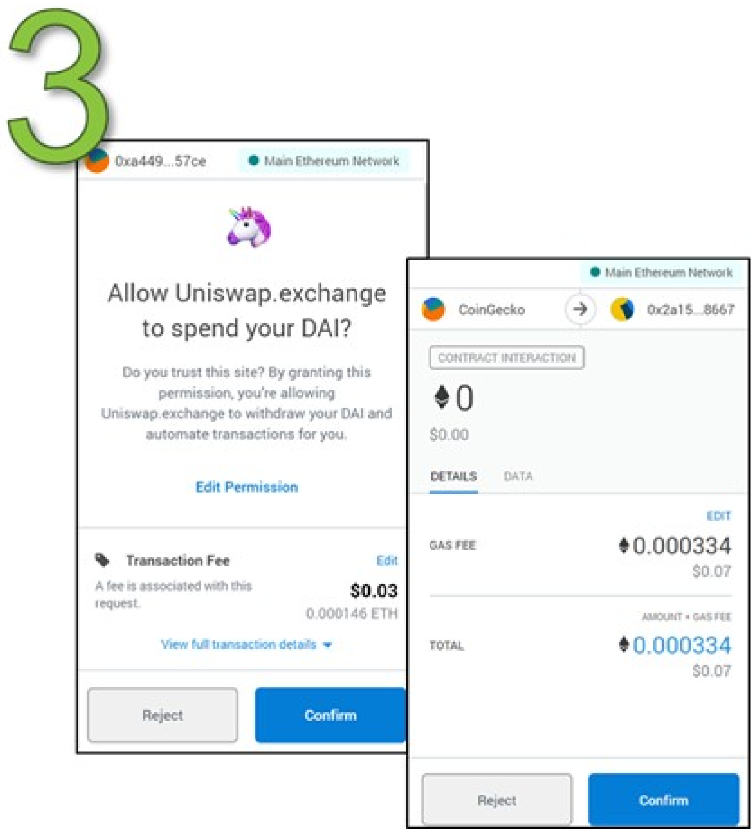
Head on over to https://uniswap.io/ and click swap token

- To start using Uniswap, you will need to connect to a wallet. You may connect your Metamask wallet. Connecting your wallet is free, all you need to do is sign a transaction



Step 2

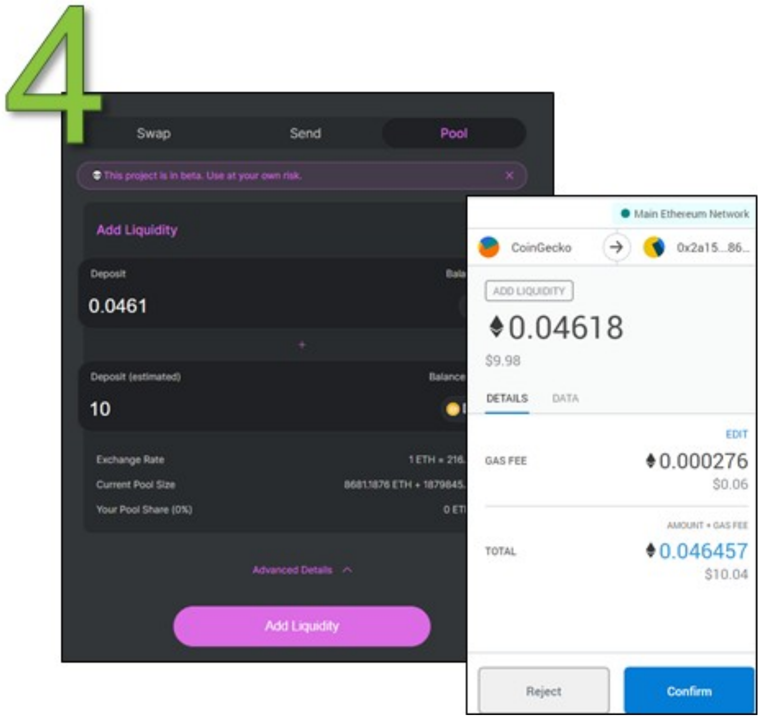
- After connecting your wallet, choose which tokens you would like to trade with, in this example, we are using DAI to buy ETH



Step 3

- If it's your first time transacting this token, you'll need to unlock it by paying a small fee
- You'll be prompted with another transaction
- Once your transaction is confirmed, you'll have your ETH!

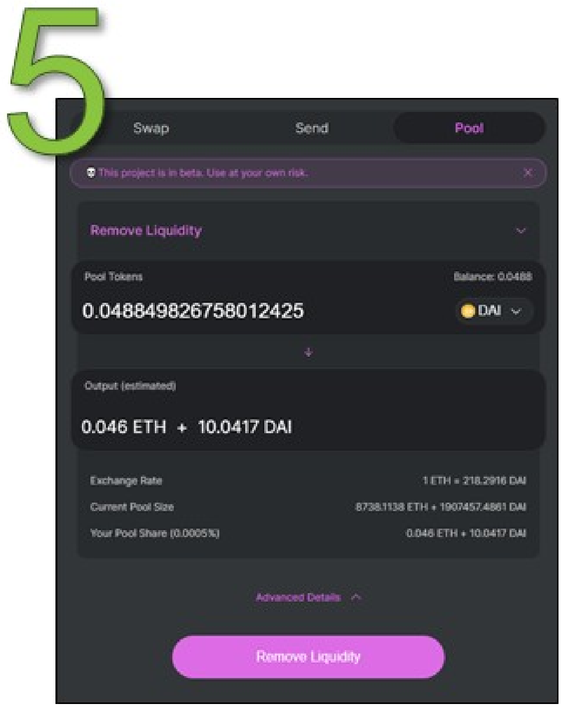
Provide Liquidity



Step 4

- Go to Pool and fill in the amount of liquidity, you wish to provide. In this case, we're providing 10 DAI worth of liquidity + 0.0461 ETH
- Note: you must have an equivalent amount of ETH to provide liquidity for that token
- After clicking add liquidity, you'll be prompted to sign another transaction
- Once that's done, you're now confirmed as a liquidity provider and stand to earn a proportionate amount of the exchange fees

Stop Providing Liquidity



Step 5

- What if you don't want to provide liquidity anymore?
- Go back to the pool and select remove liquidity
- As you can see, I'll be getting back an extra 0.0417 DAI from only 10 DAI
- Note that the ratio of my ETH and DAI is now different, so that's one of the caveats with liquidity pools, if I had removed it later, I might have a very different proportion of DAI to ETH
- Another thing to note is when removing liquidity, I trade Pool Tokens. Think of it as the proof of how much your share is in the pool. When you remove your liquidity, you will be burning the Pool tokens to get back your DAI and ETH.

~

Recommended Readings

1. Getting Started (Uniswap) https://docs.uniswap.io/
2. The Ultimate Guide to Uniswap. (DefiZap) https://defitutorials.substack.com/p/the-ultimate-guide-to-uniswap
3. A Graphical Guide for Understanding Uniswap (EthHub) https://docs.ethhub.io/guides/graphical-guide-for-understanding-uniswap

4. Uniswap — A Unique Exchange (Cyrus Younessi) https://medium.com/scalar-capital/uniswap-a-unique-exchange-f4ef44f807bf
5. What is Uniswap? A Detailed Beginner's Guide (Bisade Asolo) https://www.mycryptopedia.com/what-is-uniswap-a-detailed-beginners-guide/
6. Are Uniswap's Liquidity Pools Right for You? (Chris Blec) https://defiprime.com/uniswap-liquidity-pools
7. Understanding Uniswap Returns (Pintail) https://medium.com/@pintail/understanding-uniswap-returns-cc593f3499ef
8. UniSwap Traction Analysis (Ganesh) https://www.covalenthq.com/blog/understanding-uniswap-data-analysis/
9. A Deep Dive into Liquidity Pools (Rebecca Mqamelo) https://blog.zerion.io/liquidity-pools-8ac8cf8cf230

dYdX

dYdX is a decentralized exchange protocol for lending, borrowing and margin/leveraged trading. It currently supports 3 assets—ETH, USDC, and DAI. Through the use of off-chain order books with on-chain settlements, the dYdX protocol aims to create efficient, fair and trustless financial markets not governed by any central authority.

At first glance, dYdX appears to have some similarities to Compound - users can supply assets (lend) to earn interest and also loan assets (borrow) after depositing collateral. However, dYdX takes it one step further by incorporating a margin and leveraged exchange with ETH margin trading up to 5X leverage using either DAI or USDC.

~

Lending

ASSET	PRICE	INTEREST RATE (APR)
ETH	$280.89	Earn 0.02% / Pay 0.39%
USDC	$1.00	Earn 6.06% / Pay 7.07%
DAI	$1.00	Earn 8.35% / Pay 9.29%

If you are a crypto holder who would like to generate some passive income on your otherwise unproductive cryptoassets, you may consider lending it out on dYdX for some yield. It is relatively low risk and by depositing it into dYdX, interest accrues every second without any additional maintenance or management needed. As a lender on dYdX, you only need to be mindful of the earned Interest Rate (APR) - this represents how much you will earn from lending out your assets.

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Who pays the interest for my deposit?

The interest you earn is paid by other users who are borrowing the same asset. dYdX only allows for over-collateralized loans. This means that borrowers must always have enough collateral to pay back their loaned amount. If a borrower's collateral falls below the 115% collateral ratio threshold (i.e < $115 of ETH for $100 DAI loan), their collateral is automatically sold until they fully cover their position.

Interest rates are dynamic and change over time based on supply and demand, ensuring that users will always earn market rates. Additionally, both the initial capital and interest earned can be deposited and withdrawn at any time.

~

Borrowing

You can use dYdX to borrow any of the supported assets (ETH, DAI and USDC) as long as a 1.25x initial / 1.15x minimum collateral ratio is maintained. Borrowed funds are deposited directly to your wallet and can be freely transferred, exchanged or traded.

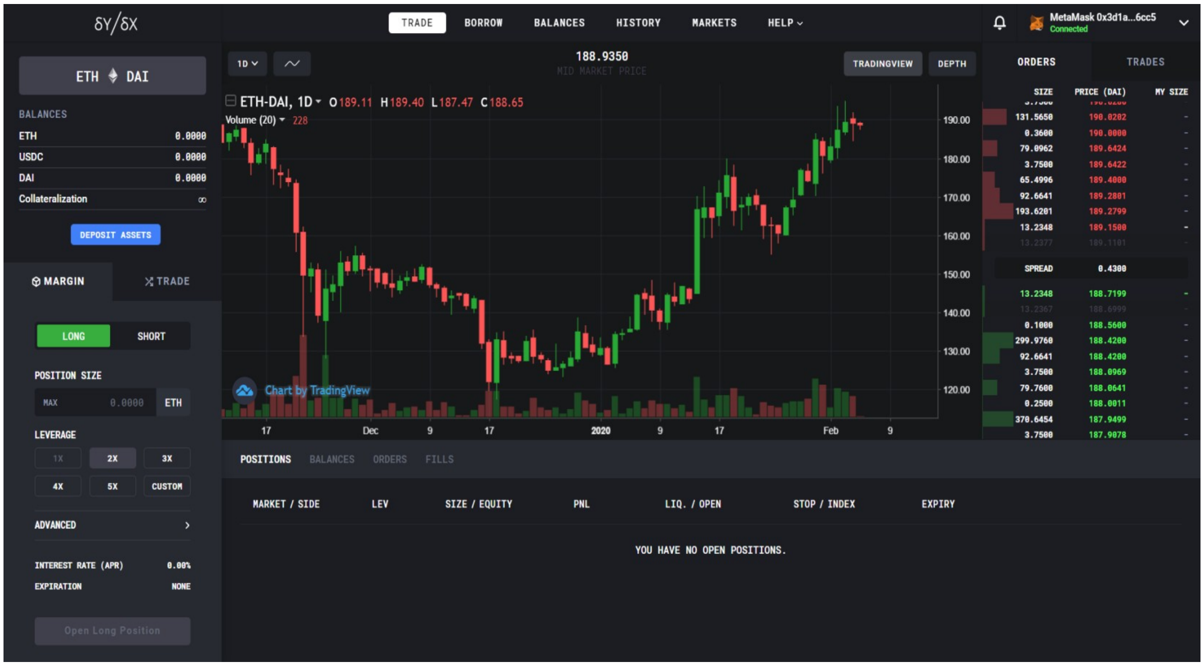
As a borrower on dYdX, the two numbers you need to look out for are:

(i) **Interest Rate (APR)** - how much you pay to loan the money

(ii) **Account collateralization ratio** - This is the ratio of your asset/loan amount. You can borrow until this ratio is 125%, and you will be liquidated once it falls below 115%.

~

Margin and Leveraged Trading



Trade Page

In dYdX you can enter either short or long positions with leverages up to 5x. When margin trading on dYdX, funds are automatically borrowed from lenders on the platform.

Consider a scenario where you start with 300 DAI & 0 ETH in your dYdX account. If you are going to short ETH (assume ETH is now $150), you will:

1. Take a loan of 1 ETH ($150)
2. Sell loaned ETH for 150 DAI, balance in dYdX is now 450 DAI & -1 ETH
3. Assuming ETH price goes to $100, you can now rebuy 1 ETH for $100 to repay your debt.
4. Your final balance is 350 DAI—you profit 50 DAI ($50)

With dYdX, you don't need to actually own ETH to enter a short position. You can borrow it and enter a short position it all in one place.

Pro Tip

Collateral used to secure margin trades continuously earns interest, meaning you don't have to worry about losing out on interest when waiting for an order *to fill. This feature is unique to dYdX as far as we know as of the time of writing.*

~

What is leverage?

Consider two different leveraged positions scenarios (numbers approximated) for a trader who has 10 ETH ($150 per ETH), or $1500. In the first scenario, the trader enters a **5x long position with 1 ETH ($150):**

1. The position size would be 5 ETH ($750)
2. 10% of the Portfolio is at risk (1/10 ETH used)
3. A price movement of ~10% ($15 on ETH) downwards will liquidate the trader's position, meaning there is very little buffer for price spikes.

On the other hand, if the trader enters a **2x long position with 1 ETH ($150)**:

1. The position size would be 2 ETH ($300)
2. 10% of the portfolio is at risk (1/10 ETH used)
3. A price movement of ~45% ($65 on ETH) downwards will liquidate the trader's position.

Essentially, leverage is really just a factor of how much risk a trader wants to take (in terms of exposure to price movements), which in turn determines a trader's distance to liquidation. High risk, high rewards!

Note: Margin positions for trades made from the US are limited to 28 days as of Feb 2020.

~

What is liquidation?

On dYdX, whenever a position falls below the collateral threshold of 115%, any existing borrows will be deemed as risky and in order to protect lenders, risky positions are liquidated. Collaterals backing the borrows will be sold until negative balances are 0, along with a liquidation fee of 5%.

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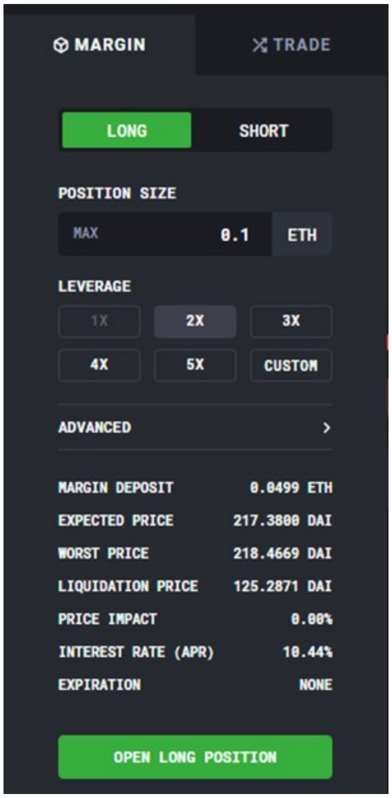
How are Profits/losses calculated?



For example, you open a 5x long with a 3 ETH deposit with an open price of $220.

You'll need to borrow $220*12 = 2640 DAI to buy 12 additional ETH (Total of 15 ETH locked in your position)

If you close the position at 250, you'll need to pay back your loan of 2640 DAI with
= 2640/250 ETH
= 10.56 ETH

This will leave you with 15 - 10.56 = 4.44 ETH. So your profit would be 4.44-3 = 1.44 ETH

Steps to calculate profit:

1. Determine initial leverage and deposit amount to get position size (Leverage*Deposit)
2. Loan Amount = (Position Size - Deposit) * Open Price
3. Loan to pay back = Loan Amount/Closing Price
4. Balance = Position Size - Loan to Pay back

5. Profit = Balance - Initial Deposit

And that's it for dYdX—if you're keen to get started or test it out, we've included step-by-step guides on how to (i) lend to earn interest, (ii) borrow and (iii) margin/leverage trade. Otherwise, head on to the next section to read more on the next DeFi app!

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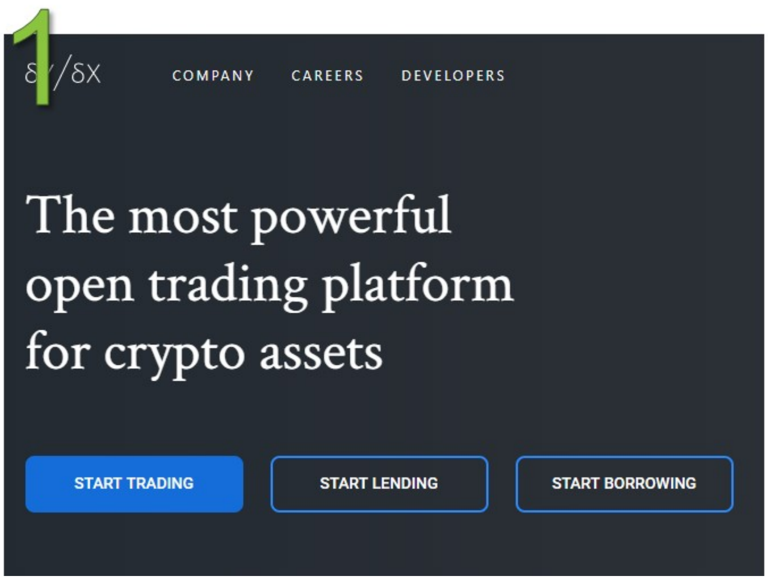
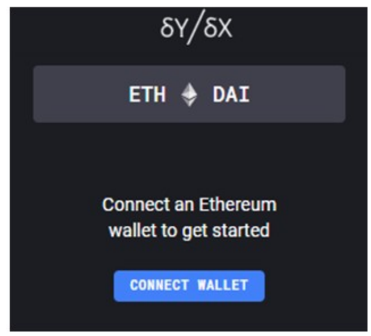
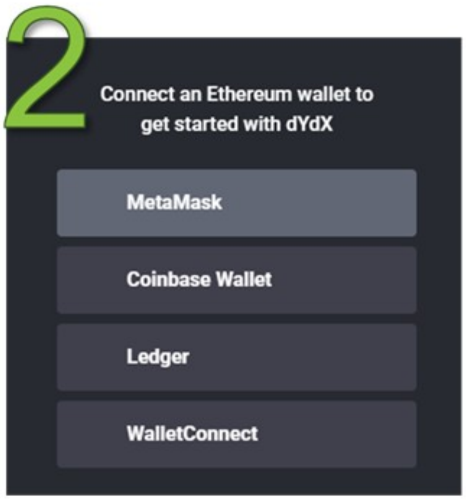
dYdX: Step-by-Step Guide

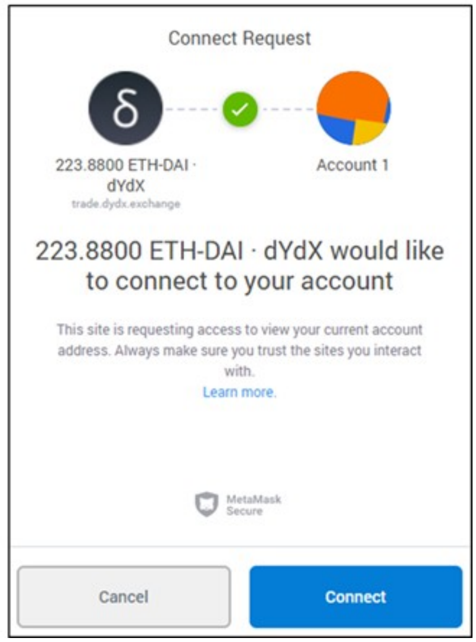




Step 1

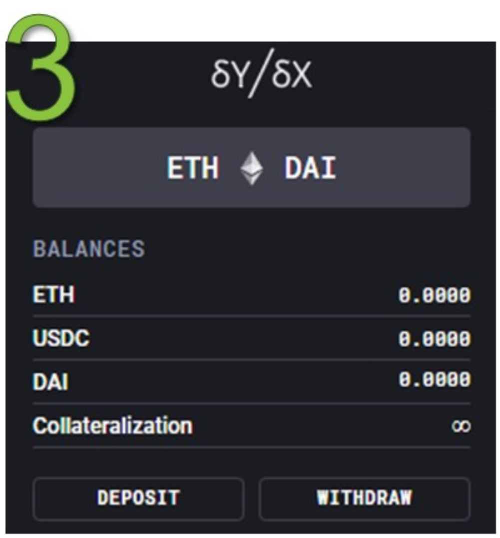
- Go to https://dydx.exchange/
- Click "START TRADING"
- Click connect wallet on the sidebar

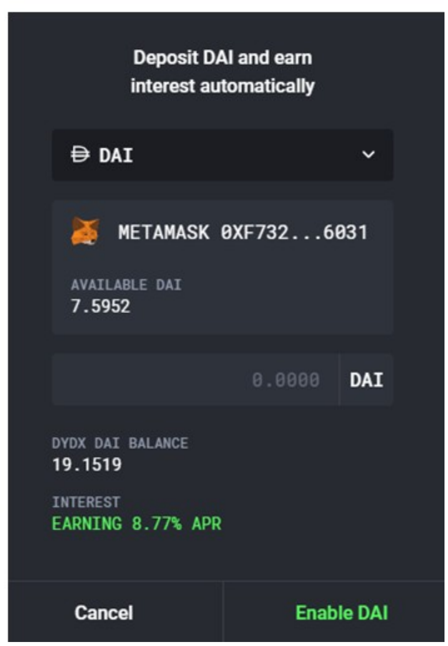




Step 2

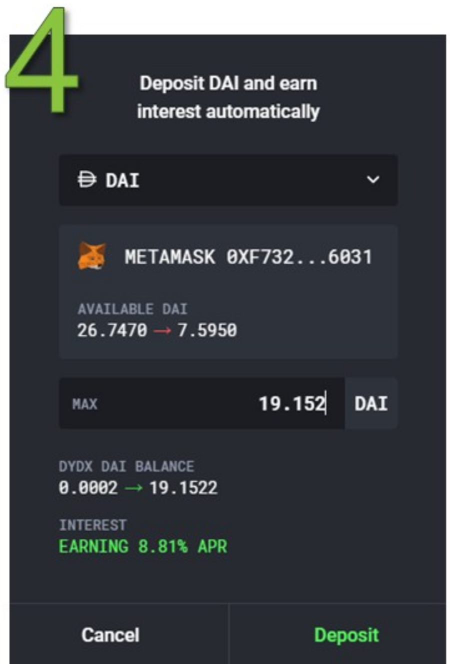
- Choose which wallet to connect

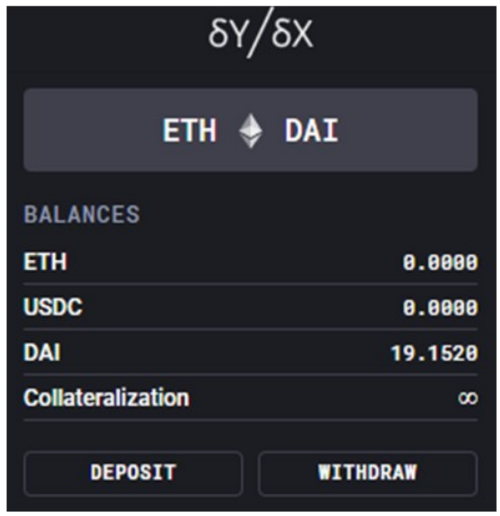




Step 3

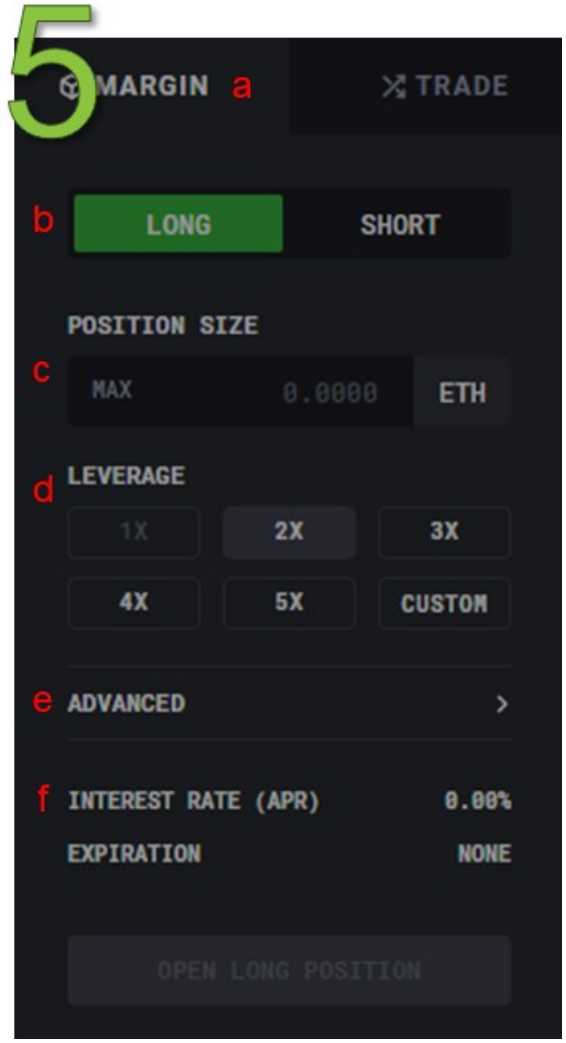
- You have no balance in your dYdX account
- Click “Deposit”
- If you are a first-timer, you will need to enable the token you wish to deposit. In this case, I wish to deposit DAI





Step 4

- Enter the amount of DAI you wish to deposit and proceed
- You will see the balance upon confirmed transaction

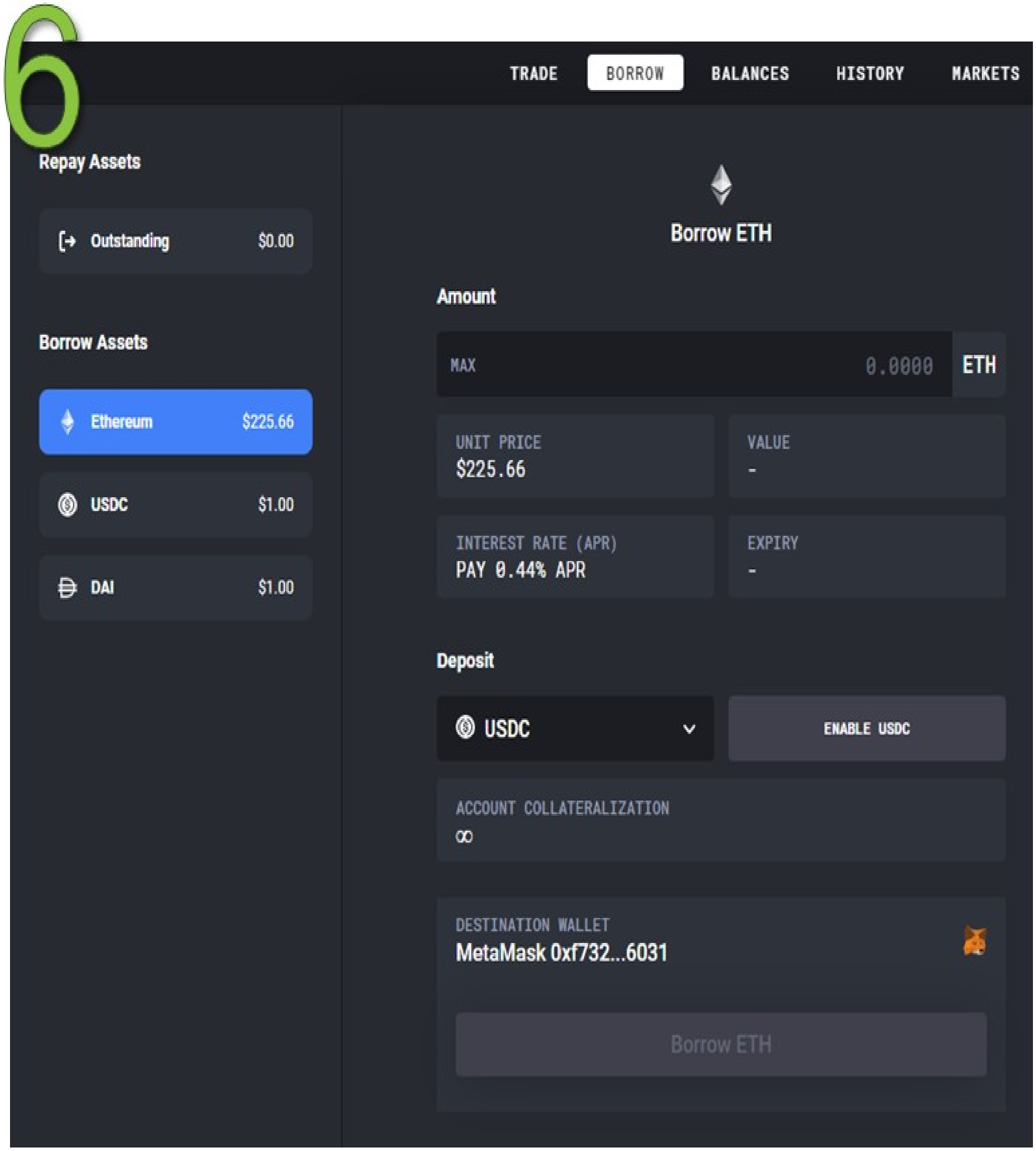


Step 5

- Now you can start trading!
- Here is a little guidance:

1. You can trade Margin or a normal spot trading.
   Margin will incur interest because you are trading on borrowed funds
2. This is the position you could take "long" or "short"
3. Position size is how much you wish to purchase to trade
4. This is the size of your borrowing. If your dYdX has 1 ETH, you could borrow up to 5X (now your position size should be up to 5 ETH)

5. This is where you would fix how much slippage you allow to happen for your position price
6. How much interest will incur you depending on the size of your margin (borrowing)



Step 6

- Alternatively, you could borrow ETH, USDC and DAI
- You will have to put up collateral before you can borrow
- You will need to enable the coin before you could start borrowing

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Recommended Readings

1. dYdX Exchange Review http://defipicks.com/2019/11/23/dydx-exchange-review/
2. Margin Trading on Centralized vs. Decentralized Exchanges (Syed Shoeb) https://medium.com/nuo-news/why-you-should-choose-decentralized-margin-trading-over-centralized-e309e61e6e72
3. Liquidators: The Secret Whales Helping DeFi Function https://medium.com/dragonfly-research/liquidators-the-secret-whales-helping-defi-function-acf132fbea5e